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Packer assembly with an anti-extrusion assembly

Abstract

A packer assembly for use within a wellbore. The packer assembly may include a mandrel, a seal assembly disposed about the mandrel, an anti-extrusion assembly disposed about the mandrel proximate the seal assembly to prevent extrusion of the seal assembly, and a deployment system. The anti-extrusion assembly may include a cradle cone that may include a plurality of arcuate surfaces on an exterior of the cradle cone and a plurality of cradle arms. Each cradle arm may include a flexible tip and be positioned on a respective arcuate surface of the cradle cone. The deployment system may be operable to compress the seal assembly such that the seal assembly seals against a wellbore wall and to rotate the plurality of cradle arms such that the flexible tips of the cradle arms contact the wellbore wall to prevent extrusion of the seal assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage entry under 35 U.S.C. 371 of International Application No. PCT/US2022/051835, entitled “PACKER ASSEMBLY WITH AN ANTI-EXTRUSION ASSEMBLY,” filed Dec. 5, 2022, which claims the benefit of Indian Patent Application number 202121057236, entitled “High Expansion Cradle Shaped Anti-Extrusion Ring and Ax Type Elastomeric Seal for Bridge Plugs, Packer, or Frac Plug,” filed Dec. 9, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(2) Hydrocarbons produced from a subterranean formation oftentimes have sand or other particulates disposed therein. As the sand is undesirable to produce, many techniques exist for reducing the sand content in the hydrocarbons. Gravel packing is one technique used to filter and separate the sand from the hydrocarbons in a wellbore. Gravel packing generally involves pumping a gravel slurry, including gravel dispersed within a carrier fluid, down a work string and into the annulus formed between a completion assembly and the wall of the wellbore. The gravel is used to filter and separate the sand from the hydrocarbons as the hydrocarbons flow from the formation, into a completion assembly, and up to the surface.

(3) One or more packers are oftentimes set or actuated prior to gravel packing. Upon actuation, the packers expand radially outward into contact with the wall of the wellbore to isolate different layers or zones of the formation. Isolating the different zones prevents the cross-flow of fluids (e.g., hydrocarbon fluids such as oil or gas) between the different zones and reduces the amount of water produced from the formation.

SUMMARY

(4) A packer assembly for use within a wellbore according to one or more embodiments of the present disclosure includes a mandrel, a seal assembly disposed about the mandrel, an anti-extrusion assembly disposed about the mandrel proximate the seal assembly to prevent extrusion of the seal assembly, and a deployment system. The anti-extrusion assembly includes a cradle cone including a plurality of arcuate surfaces on an exterior of the cradle cone and a plurality of cradle arms. Each cradle arm includes a flexible tip and be positioned on a respective arcuate surface of the cradle cone. The deployment system is operable to compress the seal assembly such that the seal assembly seals against a wellbore wall and to rotate the plurality of cradle arms such that the flexible tips of the cradle arms contact the wellbore wall to prevent extrusion of the seal assembly.

(5) A completion system according to one or more embodiments of the present disclosure includes a tubing string positionable within the wellbore and a packer assembly coupled to the tubing string. The packer assembly includes a mandrel, a seal assembly disposed about the mandrel, an anti-extrusion assembly disposed about the mandrel proximate the seal assembly to prevent extrusion of the seal assembly, and a deployment system. The anti-extrusion assembly includes a cradle cone including a plurality of arcuate surfaces on an exterior of the cradle cone and a plurality of cradle arms. Each cradle arm includes a flexible tip and be positioned on a respective arcuate surface of the cradle cone. The deployment system is operable to compress the seal assembly such that the seal assembly seals against a wellbore wall and to rotate the plurality of cradle arms such that the flexible tips of the cradle arms contact the wellbore wall to prevent extrusion of the seal assembly.

(6) A method of completing a wellbore according to one or more embodiments of the present disclosure includes positioning a packer assembly within a wellbore. The method also includes compressing a seal assembly of the packer assembly via a deployment system of the packer assembly such that the seal assembly seals against a wellbore wall. The method further includes rotating a plurality of cradle arms of an anti-extrusion assembly of the packer assembly via the deployment system such that flexible tips of the cradle arms contact the wellbore wall to prevent extrusion of the seal assembly.

(7) However, many modifications are possible without materially departing from the teachings of

this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various described technologies. The drawings are as follows:

(2) FIG. 1 is a schematic view of a well system according to one or more embodiments of the present disclosure;

(3) FIG. 2 is a packer assembly according to one or more embodiments of the present disclosure;

(4) FIG. 3 is a cross-sectional view of a sealing element of the packer assembly of FIG. 2; and

(5) FIG. 4 is a cross-sectional view of the anti-extrusion assembly of FIG. 2; and

(6) FIG. 5 is a partial view of the packer assembly of FIG. 5 in a partially deployed position.

DETAILED DESCRIPTION

(7) In the following description, numerous details are set forth to provide an understanding of some embodiments of the present disclosure. However, it will be understood by those of ordinary skill in the art that that embodiments of the present disclosure may be practiced without these details and that numerous variations or modifications from the described embodiments may be possible.

(8) In the specification and appended claims: the terms “connect,” “connection,” “connected,” “in connection with,” “connecting,” “couple,” “coupled,” “coupled with,” and “coupling” are used to mean “in direct connection with” or “in connection with via another element.” As used herein, the terms “up” and “down,” “upper” and “lower,” “upwardly” and “downwardly,” “upstream” and “downstream,” “uphole” and “downhole,” “above” and “below,” and other like terms indicating relative positions above or below a given point or element are used in this description to more clearly describe some embodiments of the disclosure.

(9) Referring now to FIG. 1, FIG. 1 is a well system **100** that includes a wellbore **102** having a deviated wellbore section **104** extending into a formation **106** containing hydrocarbon fluids. Depending on the application, the wellbore **102** may comprise one or more deviated wellbore sections **104**, e.g. horizontal wellbore sections, which may be cased or un-cased. In the example illustrated, a tubing string **108** is deployed downhole into wellbore **102** and comprises a downhole well completion system **110** deployed in the deviated, e.g. horizontal, wellbore section **104**.

(10) The downhole well completion **110** system may be constructed to facilitate production of well fluids and/or injection of fluids. By way of example, the downhole well completion **110** system may comprise at least one sand screen joint **112**, e.g. a plurality of screen assemblies **112**. Each sand screen joint **112** may include a shroud, e.g. a sand screen, **114** that cover a screen filter through which fluid may enter the corresponding sand screen joint **112** for production to a suitable location, e.g. a surface location. For example, hydrocarbon well fluids may flow from formation **106**, into wellbore **102**, and into the screen assemblies **112** via the shrouds **114**. In some embodiments, the downhole well completion system **110** also may comprise a plurality of packers **116** which may be used to isolate sections or zones **118** along the wellbore **102**.

(11) Turning now to FIG. 2, FIG. 2 is a packer assembly **216** that includes a tubular mandrel **200**, a seal assembly **202** positioned around the mandrel **200**, and anti-extrusion assemblies **204** positioned on either side of the element system. The anti-extrusion assemblies **204** are activated via a piston **206** of a deployment system **208**, which also compresses the seal assembly **202**. The seal assembly **202**, when compressed, expands in radial direction to seal an annular space between the packer

assembly **216** and a casing or a formation. The deployment system **208** can be actuated via hydrostatic pressure in the well, applying hydraulic pressure from the surface, using a mechanical tool, or any combination thereof.

(12) Turning now to FIG. **3**, FIG. **3** is an enlarged view of the seal assembly **202** of FIG. **2**. The seal assembly **202** includes multiple sealing elements **300**, **304**, **306**, **308** that are retained via two sets of foldback rings **310**, **312**. In one or more embodiments, the center sealing element **300** comprises a soft durometer elastomer (e.g., an elastomer having a durometer between approximately 60 and approximately 70) having a cylindrical groove **314** formed on the inner surface adjacent the mandrel **200**. The groove **314** is sized and positioned such that the center sealing element **300** will buckle radially outward when compression force is applied to the seal assembly **202**, as shown in FIG. **5**. An inner sealing element **304** comprises a soft durometer elastomer and is positioned within the groove **314** to reduce or prevent the formation of air or fluid pockets within the seal assembly **202** once the system is set.

(13) In one or more embodiments, wedge elements **306** comprising a hard durometer elastomer (e.g., an elastomer having a durometer of between approximately 80 and approximately 90) are positioned on either side of the center sealing element **300**. In other embodiments, the wedge elements **306** may be made of metal, polyether ether ketone (“PEEK”), or any other similar polymer. The wedge elements **306** each include an angled surface **316** on the surface opposite the center sealing element **300** and an arcuate surface **318** contacting the center sealing element **300**.

(14) A pair of end elements **308** is positioned adjacent to the angled surface **316** of the wedge elements. In one or more embodiments, the end elements **308** comprise a soft durometer elastomer to allow the end elements **308** to deform within the annular gap between the center sealing element **300** and a tubular or wellbore wall when the seal assembly **202** is compressed and create a seal. The pair of foldback rings **310**, **312** are formed from a ductile metal or polymer and partially cover portions of the end elements **308** opposite the wedge elements **306**. The foldback rings **310**, **312** may deform such that the foldback rings **310**, **312** remain in contact with the end elements **308** as the seal assembly **202** is compressed. Further, the foldback rings **310**, **312** may each include slots or gaps and be aligned with each other in such a way that one foldback ring **310**, **312** covers the slots or gaps in the adjacent foldback ring **310**, **312** to reduce or prevent extrusion of the end elements **308**.

(15) Turning now to FIG. **4**, FIG. **4** is an enlarged view of the anti-extrusion assembly **204** of FIG. **2**. The anti-extrusion assembly includes arcuate cradle arms **400** and a cradle cone **402** that includes plurality of arcuate surfaces **404** on the exterior of the cradle cone. Each cradle arm **400** includes a flexible tip **406** that can deform against a casing or wellbore wall. This allows the anti-extrusion assembly **204** to account for ovality and irregularity of the wellbore. The cradle arms **400** are each positioned an arcuate surface **404** of the cradle cone **402** and are retained on the arcuate surface **404** via guides **408** and support members **410**, such as leaf springs. Additionally, the support members **410** help to retain the cradle arms **400** in position against a casing or borehole wall once the anti-extrusion assembly **204** has been deployed, as described in more detail below. In one or more embodiments, the anti-extrusion assembly **204** also includes a foldback ring **414** to prevent the premature actuation of the anti-extrusion assembly **204**.

(16) In one embodiment, the anti-extrusion assembly **204** includes 20 cradle arms **400** positioned about a cradle cone **402** that includes twenty arcuate surfaces **404**. In other embodiments, the anti-extrusion assembly may include less than twenty cradle arms **400** and respective arcuate surfaces **404** or more than twenty cradle arms **400** and respective arcuate surfaces **404**.

(17) As discussed above, the anti-extrusion assembly **204** is deployed along with the seal assembly **202** via the deployment system **208**. When actuated via the deployment system **208**, the cradle arms **400** rotate to contact the casing or wellbore wall, as shown in FIG. **5**. Further, the deployment system **208** includes an angled or curved contact surface **412** that is positioned adjacent to the cradle arms **400** to ensure that the deployment system **208** remains in contact with the cradle arms

as the cradle arms **400** are actuated.

(18) As used herein, a range that includes the term between is intended to include the upper and lower limits of the range; e.g., between 50 and 150 includes both **50** and **150**. Additionally, the term “approximately” includes all values within 5% of the target value; e.g., approximately 100 includes all values from 95 to 105, including 95 and 105. Further, approximately between includes all values within 5% of the target value for both the upper and lower limits; e.g., approximately between 50 and 150 includes all values from 47.5 to 157.5, including 47.5 and 157.5.

(19) Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Claims

1. A packer assembly for use within a wellbore, the packer assembly comprising: a mandrel; a seal assembly disposed about the mandrel, the seal assembly including: a center sealing element; a first end element disposed on a first side of the center sealing element and a second end element disposed on a second side of the center sealing element; and a first wedge element disposed between the first end element and the center sealing element and a second wedge element disposed between the second end element and the center sealing element, each wedge element including an angled top surface; a first anti-extrusion assembly disposed about the mandrel and adjacent to the first end element and a second anti-extrusion assembly disposed about the mandrel and adjacent to the second end element, each anti-extrusion assembly comprising: a cradle cone comprising a plurality of arcuate surfaces on an exterior of the cradle cone; and a plurality of cradle arms, each cradle arm comprising a flexible tip and positioned on a respective arcuate surface of the cradle cone; and a piston disposed adjacent to the first anti-extrusion assembly and operable to: rotate the plurality of cradle arms of each anti-extrusion assembly such that the flexible tips of the cradle arms contact a wellbore wall; and compress the seal assembly such that the first end element slides along the angled top surface of the first wedge element to engage the wellbore wall, the second end element slides along the angled top surface of the second wedge element to engage the wellbore wall, and the center sealing element extends radially towards the wellbore wall.

2. The packer assembly of claim 1, wherein the anti-extrusion assembly further comprises a foldback ring.

3. The packer assembly of claim 1, wherein the anti-extrusion assembly further comprises guides.

4. The packer assembly of claim 1, wherein the anti-extrusion assembly further comprises springs.

5. The packer assembly of claim 1, wherein the center sealing element comprises a groove positioned such that the center sealing element buckles radially outward when compressed.

6. The packer assembly of claim 1, wherein the seal assembly further comprises foldback rings positioned adjacent to the end elements.

7. A completion system for use within a wellbore, the completion system comprising: a tubing string positionable within the wellbore; and a packer assembly coupled to the tubing string and comprising: a mandrel; a seal assembly disposed about the mandrel, the seal assembly including: a center sealing element; a first end element disposed on a first side of the center sealing element and a second end element disposed on a second side of the center sealing element; and a first wedge element disposed between the first end element and the center sealing element and a second wedge element disposed between the second end element and the center sealing element, each wedge element including an angled top surface; a first anti-extrusion assembly disposed about the mandrel and adjacent to the first end element and a second anti-extrusion assembly disposed about the mandrel and adjacent to the second end element, each anti-extrusion assembly comprising: a cradle cone comprising a plurality of arcuate surfaces on an exterior of the cradle cone; and a plurality of

cradle arms, each cradle arm comprising a flexible tip and positioned on a respective arcuate surface of the cradle cone; and a piston disposed adjacent the first anti-extrusion assembly and operable to: rotate the plurality of cradle arms of each anti-extrusion assembly such that the flexible tips of the cradle arms contact a wellbore wall; and compress the seal assembly such that the first end element slides along the angled top surface of the first wedge element to engage the wellbore wall, the second end element slides along the angled top surface of the second wedge element to engage the wellbore wall, and the center sealing element extends radially towards the wellbore wall.

8. The completion system of claim 7, wherein the anti-extrusion assembly further comprises a foldback ring.

9. The completion system of claim 7, wherein the anti-extrusion assembly further comprises guides.

10. The completion system of claim 7, wherein the anti-extrusion assembly further comprises springs.

11. The completion system of claim 7, wherein the center sealing element comprises a groove positioned such that the center sealing element will buckle radially outward when compressed.

12. The completion system of claim 7, wherein the seal assembly further comprises foldback rings positioned adjacent to the end elements.

13. A method of completing a wellbore, the method comprising: positioning a packer assembly within a wellbore, the packer assembly including: a seal assembly disposed about a mandrel, the seal assembly including: a center sealing element; a first end element disposed on a first side of the center sealing element and a second end element disposed on a second side of the center sealing element; and a first wedge element disposed between the first end element and the center sealing element and a second wedge element disposed between the second end element and the center sealing element, each wedge element including an angled top surface; a first anti-extrusion assembly disposed about the mandrel and adjacent to the first end element and a second anti-extrusion assembly disposed about the mandrel and adjacent to the second end element, each anti-extrusion assembly comprising: a cradle cone comprising a plurality of arcuate surfaces on an exterior of the cradle cone; and a plurality of cradle arms, each cradle arm comprising a flexible tip and positioned on a respective arcuate surface of the cradle cone; and a piston disposed adjacent the first anti-extrusion assembly and operable to: rotate the plurality of cradle arms of each anti-extrusion assembly such that the flexible tips of the cradle arms contact a wellbore wall; and compress the seal assembly such that the first end element slides along the angled top surface of the first wedge element to engage the wellbore wall, the second end element slides along the angled top surface of the second wedge element to engage the wellbore wall, and the center sealing element extends radially towards the wellbore wall; compressing the seal assembly of the packer assembly via the piston such that the seal assembly seals against the wellbore wall; and rotating the plurality of cradle arms of the first and second anti-extrusion assemblies such that flexible tips of the cradle arms contact the wellbore wall to prevent extrusion of the seal assembly.

14. The method of claim 13, wherein compressing the seal assembly further comprises compressing the seal assembly such that a center sealing element of the sealing assembly buckles radially outward.
